Income Classification & Customer Segmentation: Predictive Modeling Report for Retail Marketing

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1. Executive summary

This project focuses on two key business objectives:

1. Income Classification: Predict whether an individual earns ≤ $50K or > $50K using demographic and employment data.
2. Customer Segmentation: Identify distinct customer groups to enable targeted marketing strategies.

The dataset presents a substantial class imbalance of roughly 93.6% of the weighted population earns below $50,000, with only 6.4% in the higher-income bracket. This imbalance directly shaped modeling decisions, from sampling strategy to evaluation metrics and decision threshold selection. CatBoost was selected as the classification model given its native handling of categorical variables and strong performance under class imbalance. The segmentation model identifies 5 meaningful population clusters that differ substantially in employment status, education, age profile, and income likelihood, each offering distinct implications for targeting and messaging strategy.

2. Data Overview & Exploration

2.1 Dataset Source & Structure

The dataset is derived from the U.S. Census Bureau surveys (1994–1995) and includes:

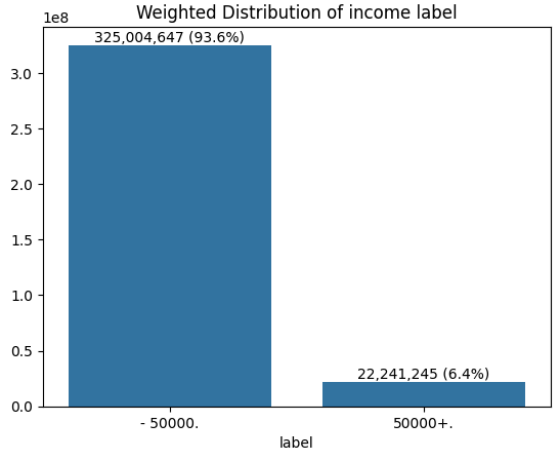
- 40 demographic & employment variables
- A binary income label (≤50K, >50K)
- A sampling weight column representing population distribution

The dataset contains 199,523 records drawn from stratified census sampling. Each record includes 40 variables spanning demographics, employment, household structure, and migration history, along with a sampling weight and a binary income label. Variables were split into 7 continuous features (including age, wage per hour, capital gains, capital losses, dividends from stocks, number of persons worked for employer, and weeks worked in year) and 33 categorical features covering attributes such as class of worker, education level, marital status, occupation, race, sex, and geographic origin.

The weight column reflects each record's proportional representation in the general population and was used throughout EDA to produce population-level distributions rather than raw sample counts.

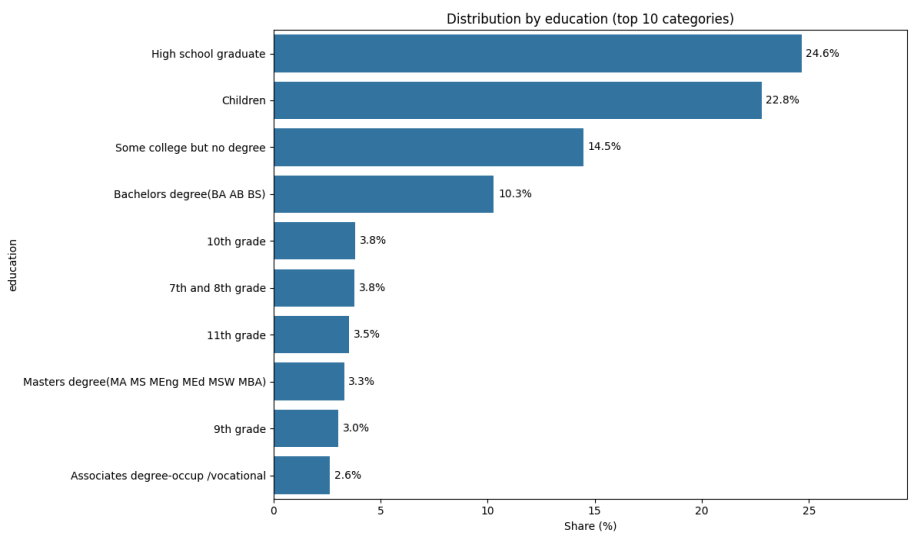
2.2 Target Variable and Class Imbalance

The target variable is severely imbalanced. On a weighted basis, the under-$50K class accounts for 325 million population units (93.6%) versus 22.2 million (6.4%) in the over-$50K class.

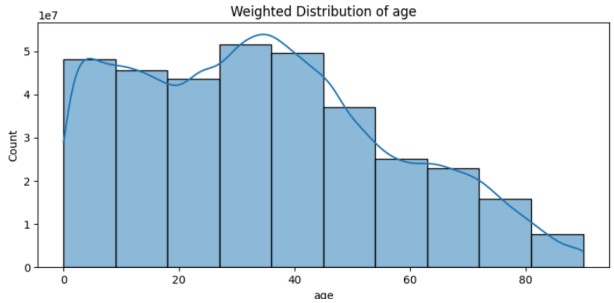


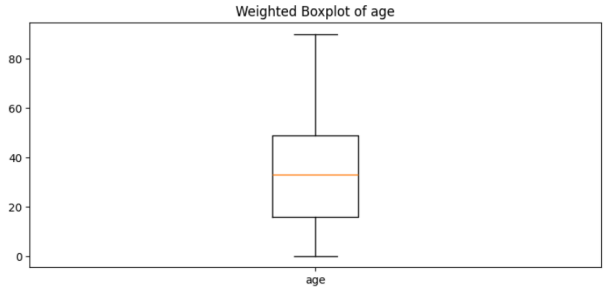
2.3 Univariate Analysis

For categorical variables, weighted horizontal bar charts were generated for each column to understand the population distribution of each category.



For continuous variables, bar plots and boxplots were built to understand spread, skew, and outlier extent. Capital gains, capital losses, and dividends from stocks showed extreme right skew with the vast majority at zero.

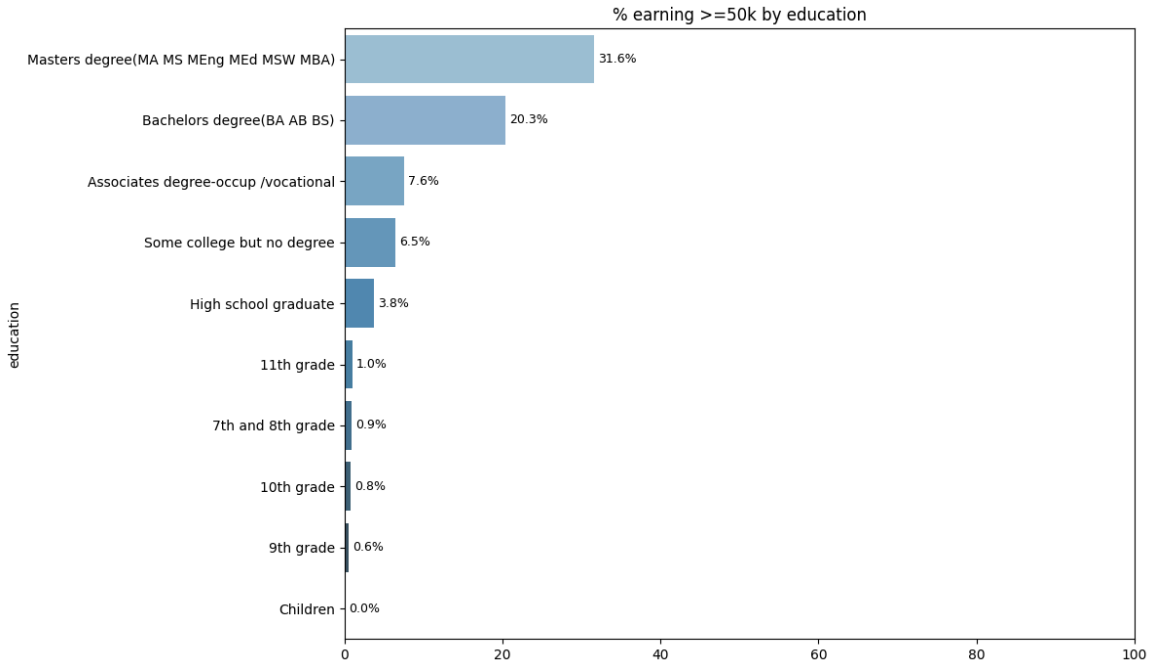




2.4 Bivariate Analysis

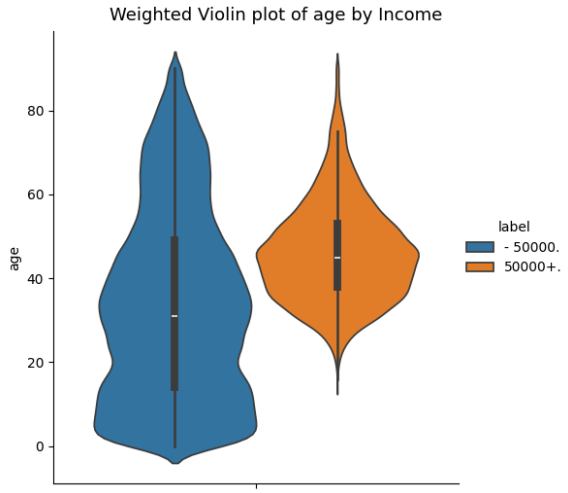
Each variable was then examined against the income label to assess predictive signal.

For categorical variables, weighted count plots split by income class revealed clear separations in features like marital status, tax filer status, class of worker, and education.
Also Percentage of high income earners were calculated as below.

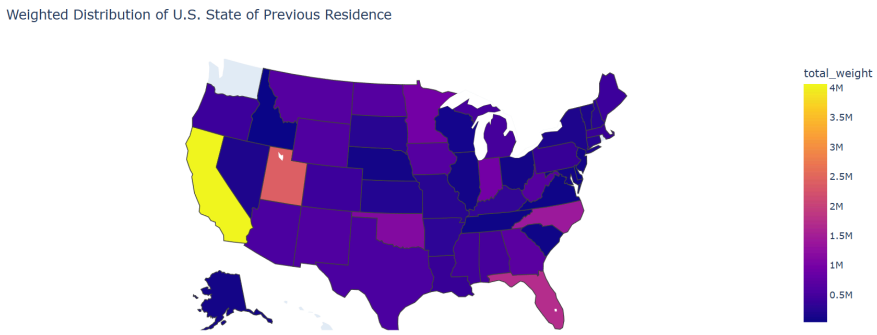


For continuous variables, weighted violin plots showed that higher-income individuals concentrated at higher ages, more weeks worked, and significantly higher capital gains and dividends.

We can see that the violin plot of age across income groups indicate that the distribution shifts towards higher age in high income

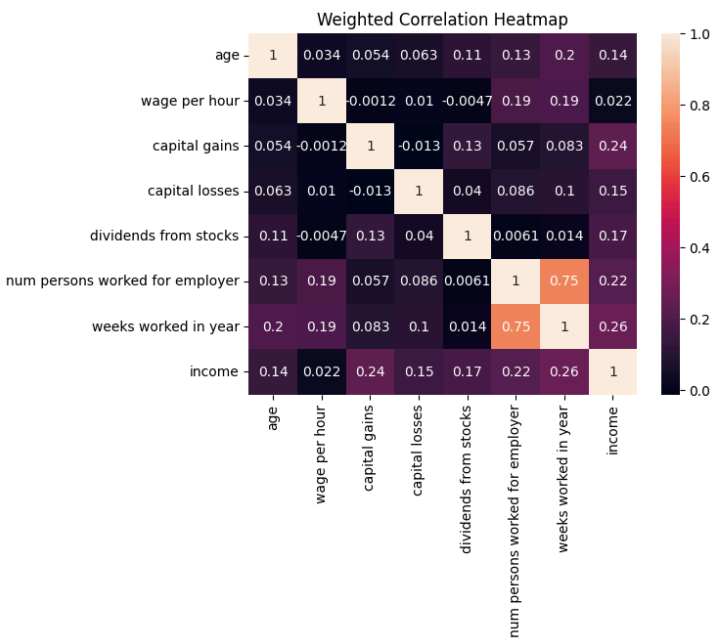


A geographic analysis was also conducted, mapping weighted population counts by state of previous residence using a Plotly choropleth to identify regional migration patterns.



2.5 Multivariate Analysis

A weighted Pearson correlation matrix was computed for all continuous features plus the binary income indicator. Weeks worked in a year shows the strongest positive correlation with income (r ≈ 0.26), followed by capital gains (r ≈ 0.24). No two continuous features are strongly collinear with each other, suggesting all can be retained without redundancy concerns.



For categorical features, one-hot encoding was applied and weighted correlations with income were computed across all binary dummies. The top predictors by absolute correlation are weeks worked in a year (0.263), executive/managerial occupation (0.242), capital gains (0.239), and detailed occupation recodes associated with professional roles.

3. Data Preprocessing

3.1 Data Cleaning

Target encoding.

The binary income label was recoded from its raw string format (- 50000. / 50000+) into a numeric indicator (0 / 1) to make it compatible with all downstream models.

Duplicate consolidation.

The raw dataset contained 199,523 rows, but a non-trivial number of records shared identical feature values across multiple survey entries.
Rather than simply dropping duplicates, rows with the same feature profile were collapsed into a single record and their population weights were summed.
This preserves the full population-representative information while reducing the working dataset to 152,896 unique records.

Missing value treatment.

Two types of missingness were addressed. The hispanic origin column contained 848 true NaN values; these were filled with "Do not know", a category that already existed in the column, to maintain consistency with the existing encoding. Eight columns (the migration-related variables and country-of-birth fields) used "?" as a placeholder for unknown responses; these were replaced with "Unknown" as an explicit category, retaining these observations in training rather than discarding them.

3.2 Feature Engineering

Education ordinal encoding.

The education column originally held 17 string categories representing discrete schooling levels. Rather than one-hot encoding this feature (which would have discarded its natural ordering), it was mapped to an integer rank from 1 (Children) to 17 (Doctorate degree), converting it into a continuous ordinal variable. This reduces dimensionality compared to dummy encoding.

Categorical type assignment.

All remaining categorical columns were cast to pandas category dtype. This enables CatBoost to handle them natively without one-hot expansion, which is memory-efficient and effective for high-cardinality features.

3.3 Train / Validation Split

The data was split into training (80%) and test (20%) sets using a stratified split on the income label. Stratification ensures that the minority class (income ≥ $50k) is proportionally represented in both sets.

3.4 Evaluation Metric Selection

The primary evaluation metric chosen is weighted PR-AUC (Precision-Recall Area Under the Curve), computed using the census population weights. This choice reflects two key factors: first, the dataset is significantly imbalanced making accuracy and ROC-AUC overly optimistic and not representative of real-world performance on the minority class. Second, in a marketing context, the cost of false positives (wasting spend on low-income individuals) and false negatives (missing genuinely high-income

prospects) are both material, and PR-AUC directly captures this precision-recall tradeoff across all thresholds.

4. Classification Model: Income Prediction

4.1 Model Selection & Rationale

Five model families were trained and evaluated on the same train/test split under identical weighted PR-AUC conditions:

Model	PR-AUC	F1	ROC-AUC
Logistic Regression (baseline)	0.6124	0.507	0.9446
Logistic Regression + PCA	0.6098	0.504	0.9440
Logistic Regression + WOE	0.5631	0.448	0.9352
Random Forest	0.6074	0.497	0.9384
XGBoost	0.6383	0.511	0.9366
LightGBM	0.6824	0.496	0.9528
CatBoost	**0.7013**	**0.601**	**0.9559**

CatBoost was selected primarily for its native handling of categorical features using ordered target statistics instead of one-hot encoding. This makes it well suited for this dataset, which contains 32 categorical variables, including several high-cardinality features such as state of previous residence and country of birth. CatBoost consistently outperformed other models on PR-AUC and achieved the highest F1 score, confirming its strength in identifying the imbalanced positive class. LightGBM showed competitive AUC, but its higher recall came at the cost of more false positives, resulting in lower precision. This tradeoff is less suitable for a marketing use case where targeting quality is as important as reach.

4.2 Feature Selection

An initial CatBoost model was trained on all 40 features. Feature importances were extracted and sorted by their cumulative contribution to model predictions: 24 features accounted for 95% of total importance, and the remaining 16 features contributed negligible signal. The reduced feature set was adopted for all subsequent experiments. Retaining only these 24 features slightly improved validation PR-AUC (0.7013 to 0.7066) while reducing training time.

4.3 Hyperparameter Tuning

Bayesian hyperparameter search was conducted using TPE algorithm over 30 evaluation trials, with early stopping (100 rounds without improvement) applied at each trial to control training time given GPU and

time constraints. The best configuration found was depth = 6, learning rate ~ 0.061, L2 leaf regularization ~ 4.23, bagging temperature ~ 0.57, and random strength ~ 0.79, yielding a final validation PR-AUC of 0.7074. The improvement from tuning was modest given the run was for only 30 trials given time and gpu constraints.

Stability of the best model was confirmed via 3-fold cross-validation on the training set, which produced a mean PR-AUC of 0.6862 with a standard deviation of only 0.0028, demonstrating low variance and no overfitting to the held-out test set.

4.4 Threshold Tuning

The default classification threshold of 0.5 is suboptimal for imbalanced datasets: at this threshold, the model achieves a recall of only 0.49 for the positive class, meaning it misses more than half of genuinely high-income individuals. Precision-recall-F1 curves were computed across all thresholds, and the threshold maximizing weighted F1 was identified at **0.3377**. At this operating point:

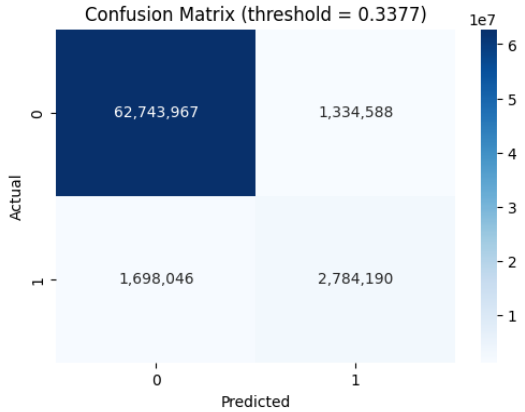
Threshold	Precision	Recall	F1
0.50 (default)	0.77	0.49	0.60
0.3377 (tuned)	**0.68**	**0.62**	**0.65**

The tuned threshold accepts a modest reduction in precision (77% to 68%) in exchange for a substantial gain in recall (49% to 62%), resulting in a net F1 improvement of 0.05.

4.5 Evaluation Results

The final tuned model evaluated at threshold 0.3377 achieves the following weighted performance on the held-out test set:

- PR-AUC: 0.7074
- ROC-AUC: 0.9569
- F1 (positive class): 0.65
- Precision (positive class): 0.68
- Recall (positive class): 0.62
- Overall accuracy: 96%



The confusion matrix shows that approximately 2.8M population-weighted individuals who earn ≥$50k are correctly identified (true positives), while approximately 1.7M are missed (false negatives) and approximately 1.3M lower-income individuals are incorrectly flagged (false positives).

SHAP analysis confirms the model's learned logic is intuitive and directionally consistent with domain expectations: higher age, higher education rank, more weeks worked, and greater capital gains and dividends all push predictions toward ≥$50k. Being enrolled in college, classified under "Not in universe" occupational codes, or being female reduces predicted income probability.

4.6 Fairness Analysis

A fairness audit was conducted across sex and race groups by computing group-level PR-AUC, false negative rates (FNR), and false positive rates (FPR) at the tuned threshold.

By sex: Male observations have a PR-AUC of 0.753 and an FNR of 0.325. Female observations show a substantially lower PR-AUC of 0.515 and an FNR of 0.595, meaning the model misses nearly 60% of high-earning women. This disparity is traceable to base rate differences in the training data (only 3.4% of women in the training set earned above $50k versus 13.1% of men) leaving the model with very limited signal on high-earning women.

By race: White individuals have the highest PR-AUC (0.718) and a moderate FNR (0.373). Black individuals (PR-AUC 0.503, FNR 0.565) and Native American individuals (PR-AUC 0.515, FNR 0.778) show the largest disparities, again reflecting low positive rates and smaller sample sizes in those groups in the underlying census data.

These disparities reflect historical income inequality in 1994–1995 U.S. society rather than a flaw in the modeling approach.

5. Segmentation Model - Customer Groups

5.1 Approach and Algorithm Choice

The segmentation model was built as a downstream layer on top of the classification pipeline. After running the trained CatBoost classifier on the full pre-processed dataset, each record was enriched with an income_prediction_score (a continuous probability representing the model's estimated likelihood of the individual earning above $50k). This score was then included as an input feature for clustering, ensuring that the resulting segments capture not just demographic and employment patterns but also predicted income tier.

K-Means was selected as the clustering algorithm. It is well-suited to this dataset given its scale (152,896 records, 36 features post-encoding). A Gaussian Mixture Model (GMM) was also evaluated as an alternative; while GMM produced lower BIC and AIC scores at higher component counts, its silhouette scores on a held-out sample were consistently below those of K-Means across all values of k tested, making K-Means the stronger performer on cluster cohesion.

5.2 Feature Engineering for Clustering

The feature set for clustering was constructed separately from the classification pipeline to reflect marketing-relevant dimensions rather than raw predictive power. Several transformations were applied: Derived features were introduced. Continuous features with heavily skewed distributions were converted to binary flags. High-cardinality categorical variables were consolidated into meaningful groups before encoding.

A Variance Inflation Factor (VIF) analysis was run to check for multicollinearity.

5.3 Dimensionality Reduction and Choosing K

All features were standardized before clustering. PCA was then applied, retaining components that explain 95% of cumulative variance, which reduced the feature space from 36 to 23 principal components. This step reduces noise from correlated features and lowers the computational cost.

The elbow method and silhouette scores were computed for k = 2 through 10. The elbow curve showed diminishing inertia reductions beyond k = 5. Silhouette scores peaked at k = 5 (0.163), with no meaningful improvement at higher values. K = 5 was selected as the final number of segments, balancing interpretability with cluster quality.

5.4 Segment Profiles

Segment 0: Dependents / Children (28%). Average age of 9, with 88% classified as "Child" in family status and 98% not working. These are minors with virtually no income signal.

Segment 1: Core Working Families (22%). Average age of 41, with 98% employed and 99% filing jointly. Most are married wage earners in dual-role households. Despite being fully employed, the low income prediction score (0.09) suggests the majority earn under $50k.

Segment 2: Independent Workers (20%). Average age of 33, with 98% working and 96% filing as single. Over half fall under "Other" family status, suggesting they are unmarried, not heading a household, and likely in the early stages of their career.

Segment 3: Older Non-Working Adults (20%). Average age of 59, with only 7% still employed. Filing status and household roles are mixed, pointing to a retirement or near-retirement population, some of whom remain married to working spouses.

Segment 4: High Income Group (9%). Average age of 46, with 97% employed, 79% receiving dividend income, high average education level (12.65) and nearly all reporting capital gains. Predominantly married, male, and highly educated, this segment carries the highest income prediction score (0.38)

5.5 Business Interpretation and Marketing Use

Segment 4 is the primary high-value target for products and services aimed at upper-income households: wealth management, premium credit products, luxury goods, and high-end home or auto financing. The combination of capital income, high education, and a strong income prediction score makes this the most reliable segment for above-$50k outreach.

Segment 1 represents the broadest reachable working population. While most earn below $50k, they are stable, employed, and household-focused. They are well-suited for mid-range financial products, family-oriented services, and aspirational retail. The classifier score can be used within this segment to identify the subset most likely to be near or above the income threshold.

Segment 2 targets younger, independent earners who are likely to be growing in income. This group responds well to products associated with life transitions: first credit cards, renter's insurance, entry-level investment accounts, and career-adjacent services. Their lower current income score does not preclude future value as customers.

Segment 3 calls for a retirement-oriented approach. Passive income, near-retirement age, and mixed employment suggest appetite for fixed-income products, healthcare-adjacent services, travel, and legacy planning.

Segment 0 (children and dependents) is not a direct marketing target for income-based products.

6. Business Recommendations

6.1 Using the Classifier in Practice

The model should be deployed as a prospect scoring tool, outputting a probability. The default threshold of 0.3377 provides a balanced tradeoff (Precision: 68%, Recall: 62%), but this should not be fixed.

Threshold selection must be driven by business economics:

- Lower threshold for higher reach, more false positives (use when marketing cost is low)
- Higher threshold for higher precision, fewer but more qualified leads (use when budget is constrained)

The model can also be extended to cost-sensitive training, where misclassification penalties are aligned with campaign costs to optimize ROI directly.

6.3 Limitations and Risk Flags

The model reflects biases present in the historical data and shows weaker performance for certain demographic groups, including women and minority populations. This creates a risk of under-targeting.

To mitigate this:

- Monitor model performance across subgroups
- Adjust thresholds or apply group-aware calibration
- Incorporate additional signals where fairness is critical

6.4 Next Steps

- Retrain the model on recent, client-specific data to improve relevance and fairness
- Incorporate cost-based optimization into model training and threshold selection
- Track campaign outcomes to continuously refine both segmentation and targeting strategy

7. References

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